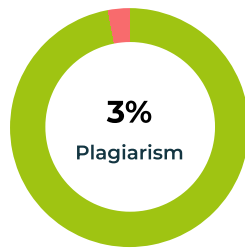


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01 [None](#)

Content

- A thorough comparative study of Logistic Regression and Support Vector Classifier (SVC) for predicting 22 crop classes using a balanced agronomic dataset.
- Evidence that the SVC model with an RBF kernel reaches a test accuracy of 98.18%, surpassing Logistic Regression (97.05%) across all macro-averaged metrics.
- The design and deployment of a complete crop and fertilizer recommendation system using Flask, which is accessible through a web browser without needing any special hardware.
- A detailed system architecture that includes data ingestion, preprocessing, model inference, fertilizer mapping, and user interface delivery.

II. RELATED WORK

The machine learning research field maintains continuous interest in precision agriculture as a subject of study. The agricultural ML survey conducted by Liakos et al. [2] identified crop management, water management and yield prediction as the three main application domains of machine learning in agriculture. The study reviewed 40 research articles which showed that neural networks and ensemble methods dominate yield prediction while basic classifiers show similar performance in crop recommendation tasks.

The AgroConsultant crop recommendation system

developed by Doshi et al. [3] uses Random Forest to achieve about 99% accuracy on soil-climate data. The model suffered from excessive tree count which resulted in delays that made it impractical for use in real-time mobile applications. The study reported accuracy metrics based on a single train-test split which lacked cross-validation and this method potentially resulted in exaggerated generalisation performance measurements. Karthik et al. [4] found that Naïve Bayes and Decision Tree classifiers produced crop recommendation results with accuracy scores between 85 and 92 percent. The research demonstrated that probabilistic models lose their ability to predict accurately when

researchers use correlated soil data which violates the assumption of feature independence between variables.

The researchers also investigated deep learning methods.

Raut and Bhagat [5] used a Convolutional Neural Network (CNN) to identify plant diseases and their system achieved 96.2% accuracy on the image data. CNN architectures provide excellent performance on visual tasks but they fail to handle tabular agricultural data while consuming much more computational power than traditional ML methods.

Nevavathi and Sivakumar [6] achieved 94.5% accuracy of their system through LSTM networks for forecasting crop yields through time series data. LSTMs need temporal sequences of historical data which smallholder farmers in developing countries cannot access.

Kour and Arora [7] achieved 96.8% accuracy in crop classification through their use of Support Vector Machines on satellite remote sensing data. The research confirmed that kernel-based SVM classifiers function effectively in high-dimensional agricultural feature spaces which led us to choose SVC for our current study.

The researchers conducted complete assessments of precision agriculture systems which identified a permanent barrier that prevents direct conversion of research prototypes into operational systems that farmers can use. The majority of published systems operate in isolation—predicting crop or fertilizer independently—without a unified recommendation pipeline. AgriVision directly addresses this architectural limitation.

III. DATASET AND EXPLORATORY DATA ANALYSIS

A. Dataset Description

The research employs the Crop Recommendation Dataset which researchers can access from Kaggle [10]. The dataset consists of 2,200 records which agricultural scientists created through their research process to establish a balanced distribution across all crop classes. The dataset consists of seven continuous input features and one categorical target variable (crop label). Table I presents all feature specifications which include biological significance, value ranges and measurement units.

TABLE I. DATASET FEATURE SPECIFICATION

Feature Description	Range / Unit	Role
Nitrogen (N) Soil macronutrient – promotes leaf/stem growth	0 – 140 mg/kg	Input
Phosphorus (P) Soil macronutrient – promotes root/flower development	5 – 145 mg/kg	Input
Potassium (K) Soil macronutrient – disease resistance & water regulation	5 – 205 mg/kg	Input
Temperature Ambient temperature governing crop metabolism		

8°C – 44°C Input
 Humidity Relative atmospheric moisture affecting transpiration
 14% – 100% Input
 pH Soil acidity/alkalinity influencing nutrient availability
 3.5 – 9.9 Input
 Rainfall Annual precipitation determining water availability
 20 – 299 mm Input
 Crop Label Recommended crop class (target variable) 22 categories Target
 Complete description of the seven input features and target label in the Kaggle Crop Recommendation Dataset.

B. Crop Class Distribution

The 22 crop classes include all major Indian and South-Asian agricultural categories because they include food grains, pulses, oilseeds, and fruit crops. List in Bold displays the complete distribution of crop classes. The class balance achieves 1. 100 samples for each class which results in 2,200 total samples to prevent majority-class bias from inflating accuracy while macro-averaged metrics show exact per-class evaluation results.
 Rice, Maize, Chickpea, Kidneybeans, Pigeonpeas, Mothbeans, Mungbean, Apple, Blackgram, Lentil, Pomegranate, Banana, Mango, Grapes, Watermelon, Muskmelon, Orange, Papaya, Coconut, Cotton, Jute, Coffee

Complete list of 22 crop classes. Each class contains exactly 100 samples.

C. Exploratory Data Analysis (EDA)

Researchers completed their exploratory analysis work before starting the modeling process. Key observations:

- Nitrogen (N) shows the highest inter-class difference because it effectively distinguishes between cereal crops (rice, maize) and legumes (chickpea, lentil).
- pH values exist in two separate bands: slightly acidic (5.5–6.5) for tropical crops (banana, mango) and near-neutral (6.5–7.5) for grain crops.
- Rainfall and humidity show a strong positive correlation (Pearson $r = 0.78$) which indicates their connection with climatic conditions.
- Temperature ranges create two distinct crop groups: tropical crops requiring temperatures above 28°C average while subtropical crops need temperatures below 22°C average.

- No missing values or duplicate records were found. The seven features showed restricted distribution patterns with no extreme outliers needing to be capped.

IV. METHODOLOGY

A. Data Preprocessing Pipeline

The three-stage preprocessing pipeline was constructed with scikit-learn through three separate stages.

1) Feature Scaling

StandardScaler was applied to all seven input features, transforming each to zero mean and unit variance: $x' = (x - \mu) / \sigma$. The SVC model requires standardisation because the RBF kernel calculates Euclidean distances between data points; unscaled features with large magnitude ranges (e.g., Rainfall: 20–299 vs pH: 3.5–9.9) would otherwise dominate the distance computation and bias the decision boundary. Standardisation also helps Logistic Regression achieve faster convergence rates.

References